MILITARY SPECIFICATION

RAWIN SET AN/GMD-1( )

(UNITS OF)

This specification is mandatory for use by all Department and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. - This specification covers the requirements for Rawin Set AN/GMD-1( ), and certain units and assemblies thereof, as listed below:

Antenna AS-462( )/GMD-1	Case CY-734( )/GMD-1
Antenna Control C-578( )/GMD-1	Case CY-735( )/GMD-1
Control Recorder C-577( )/GMD-1	Case CY-736( )/GMD-1
Pedestal AB-159( )/GMD-1	Case CY-737( )/GMD-1
Rawin Receiver R-301( )/GMD-1	Case CY-1157( )/GMD-1
	Case CY-1895( )/GMD-1

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

T-T-871	Twine, Cotton, Wrapping
QQ-S-781	Strapping, Flat; Steel
UU-P-268	Paper, Kraft, Untreated, Wrapping
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner
PPP-B-636	Box, Fiberboard
PPP-F-320	Fiberboard, Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes
PPP-P-291	Paperboard, Wrapping, Cushioning
PPP-T-60	Tape; Pressure-Sensitive Adhesive, Waterproof for packaging
PPP-T-76	Tape, Pressure-Sensitive Adhesive Paper, Water Resistant

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SPECIFICATIONS

MILITARY

MIL-E-75	Electronics, Tubes, Packaging, Packing, and Container Marking for General Specification For
MIL-P-116	Preservation, Methods of
MIL-B-117	Bag and Sleeve, Interior Packaging
MIL-L-10547	Liners, Case, and Sheet, Overwrap, Water-vaporproofed or Waterproof, Flexible
MIL-P-11268	Parts, Materials, and Processes used in Signal Corps Equipment
MIL-M-13231	Marking of Electronic Items
MIL-F-14072	Finishes for Ground Signal Equipment
MIL-C-45662	Calibration of Standards

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-169	Extreme-Temperature Cycle
MIL-STD-170	Moisture Resistance Test Cycle for Ground Signal Equipment
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-810	Environmental Test Methods for Aerospace and Ground Equipment

DRAWINGS

ELECTRONICS COMMAND

SC-DL-93528	Rawin Set AN/GMD-1( )

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 Description. - Rawin Set AN/GML-1( ) is a transportable radio direction finder designed to automatically track balloon-borne radiosondes. Antenna elevation and azimuth angles are utilized to compute wind direction and speed at various altitudes. The set also receives and delivers to a supplementary recorder intelligence of meteorological conditions as transmitted by the radiosonde.

3.2 Preproduction samples. - The contractor shall furnish three preproduction samples for approval, if required by the invitation for bids and contract. When preproduction samples are not specified by contract or purchase order three production units of the first ten (10) production units produced shall be tested to the same requirements as specified for preproduction samples under 4.3. (See 6.4)

3.2.1 Fabrication and assembly. - Preproduction samples shall be fabricated by the contractor and shall meet all requirements specified for the equipment on contract. The preproduction samples shall be assembled using parts, materials, and processes that will be employed in production, and shall be fabricated and assembled in a manner similar to that to be used in production.

3.2.2 Approval and use. - Approval of preproduction samples shall not be construed as a waiver of any specified requirement. After being released to the contractor (see 3.2.2), preproduction samples to be offered as units on contract shall be refurbished by the contractor if necessary to meet specified requirements.

3.2.3 Reference standards. - After preproduction samples have been approved, they shall be kept intact in custody of the government inspector until released by him. They shall be used as reference standards to resolve any differences of opinion regarding interpretation of requirements.

3.3 Construction. - The equipment shall be constructed in accordance with the requirements of this specification and Drawing List SC-DL-93528. The requirements of MIL-P-11268 shall apply when the part, material or process is not specified by drawings.

3.3.1 Cases. - All cases listed under 1.1 shall conform to the tests of 4.18 and all other applicable paragraphs of this specification.

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3.4 Finish. - The equipment shall be finished in accordance with MIL-F-14072 and the equipment drawings. (See 4.4)

3.5 Marking. -

3.5.1 General. - Marking shall conform to MIL-M-13231 and the equipment drawings. (See 4.4)

3.5.2 Serial Numbers. - The units listed in 1.1 shall have serial numbers except the cases.

3.6 Interchangeability. - Like units, assemblies, subassemblies and replaceable parts shall conform to requirement 7 of MIL-STD-454. (See 4.7.1)

3.7 Tropicalization of material. - In addition to the requirements of MIL-P-11268 the equipment shall be exposed to fungus and salt fog tests.

3.7.1 Fungus. - The equipment (including accessories) shall show no evidence of viable fungus or corrosion when subjected to the test specified in 4.20.1. Corrosion is any visible degradation that can be attributed to flaky, pitted, blistered, or otherwise loosened finish or metal surface.

3.7.2 Salt Fog. - The equipment and assemblies shall be exposed, in their operating configuration, to 48 hours of salt fog in accordance with paragraph 4.20.2; and shall show no evidence of corrosion as evidenced by any visible degradation that can be attributed to flaky, pitted, blistered or otherwise loosened finish or metal surface.

3.8 Performance. - The AN/GMD-1( ) shall meet all requirements of this specification with an input voltage of 105 thru 129 volts AC at 50 thru 65 Hz.

3.8.1 Tracking. - Tracking shall conform to the following:

3.8.1.1 Changes in tuning, field strength, polarization and carrier frequency. - When the receiver, tuning, signal field strength, polarization and carrier frequency are varied in accordance with 4.10, the individual bearings shall remain within 0.05° of the original target bearing.

3.8.1.2 Azimuth - elevation automatic reset. - The average reset error of five test runs as specified in 4.10.1 shall not exceed 0.02°.

3.8.1.3 Speed. - Automatic tracking in both elevation and azimuth at the antenna shall range up to 1 r.p.m.

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3.8.1.4 Travel. - The mechanical limiting devices shall limit the travel of the antenna in the vertical plane to the range of -5 to +95 degrees. The electrical limit switches on the elevation unit shall be -3(+0 -0.5) degrees and +90(+0.5-0.0) degrees.

3.8.1.5 Azimuth. - Elevation Manual Control. - The antenna shall be stationary when the azimuth and elevation antenna controls are in the center position and when positioned "on", shall have variable slewing rates up to speed specified in 3.8.1.3.

3.8.1.6 Elevation manual control. - The control shall meet the elevation limits of 3.8.1.4.

3.8.2 Mechanical. - Average deviation shall not exceed 0.03° of the original target bearing.

3.8.3 Control Recorder C-577( )/GMD-1. - The Control Recorder chart operating at speeds of 1, 2 and 10 printings per minute shall not tear the chart and the print out data (azimuth, elevation, time and asterick) shall be clear. The printed data and visual indication of the azimuth and elevation bearings of the antenna system shall be accurate to the nearest 0.05 of a degree and the time to 0.1 minute or less.

3.8.4 Antenna AS-462( )/GMD-1. - The antenna system shall conform to the following at 1680 MHz:

(a) The effective area of the antenna shall be a minimum of 1.5 square yards and the lobe width shall be 6.5° nominal measured between points 3 db down from maximum.

(b) The secondary lobes shall be attenuated a minimum of 20 db.

(c) The cross over point of the lobes shall be at a point between 0.85 and 0.95 of the lobe voltage maximum.

3.8.5 Rawin Receiver R-301( )/GMD-1. - The receiver shall conform to the following:

3.8.5.1 Frequency. - The receiver shall operate over the frequency range of 1,660 MHz or below thru 1700 MHz or above.

3.8.5.2 IF amplifier. - The two bandwidths of the IF amplifier shall be 1.5±0.3 and 0.85±0.15MHz. Each of these shall contain AM and FM outputs.

3.8.5.3 Automatic frequency control (AFC). - The corrected intermediate frequency shall lock-in from a minimum of ±1.0 MHz from center of the IF at any input level of 5 micro-volts and it shall not be more than 100 kHz in error at any signal level above 5 microvolts.

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3.8.5.4 Automatic gain control (AGC). - The ratio of the receiver output shall not exceed 8:1 when the input to the receiver is varied from 5 thru 50,000 microvolts. The AGC time constant shall not exceed 0.1 second.

3.8.5.5 Audio channel phase shift. - The phase shift in the error signal between the input to the IF amplifier and an input to the antenna control shall not be more than 20° with a maximum change in power input frequency or signal level. Any combination of line frequency and signal level conditions shall not produce a change in phase shift in excess of 40°.

3.8.5.6 Signal-to-noise ratio. - The signal-to-noise ratio when measured as shown in 4.13.6 shall not exceed 15 db.

3.8.5.7 Radiosonde pulse channel. - The receiver shall pass pulses of 40 to 125 microseconds duration at the input with the repetition rates varying from 10 to 225 per second. The output of the receiver shall be negative going pulses of 1,200 to 2,500 microseconds duration with amplitudes of 25 to 60 volts thru 500 feet of connecting cable. The receiver shall accept radiosonde data down to the point where pulse peaks are two times noise peaks measured at the IF output. There shall be no interference with radiosonde reception by pulse signals of either positive or negative polarity having a duration of less than 20 microseconds.

3.8.5.8 Audio monitor. - The audio monitor shall have an output of 5 milliwatts maximum at the minimum position of the control and 200 milliwatts minimum at the maximum position of the control measured at 200 Hz.

3.8.5.9 Microphonics. - The signal shall not disappear nor shall there be excessive jumbping of the signal, modulated or unmodulated.

3.8.5.10 Mixer, impedance. - The mixer at a nominal impedance level of 50 ohms shall not have a matching loss exceeding 1 db over the specified frequency range.

3.9 Flight test. - The antenna shall remain on target to within an average arithmetic accuracy of 0.05° in both azimuth and elevation when the elevation angles lie between 10° and 60°.

3.10 Service conditions. - The equipment shall meet the following service conditions - - where a test is referenced, meeting the test shall be considered as compliance with the requirement.

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3.10.1 Operation. - Continuous use for a period of 1 year, under the environmental requirements specified in 3.10.2 through 3.10.6, with a duty cycle of 23 hours on and 1 hour off with no more than normal maintenance and replacement of parts. Where the contractor is required to make a selection of parts, materials, processes, construction methods, etc., he shall be guided by this requirement and 6.2. Approval of the preproduction sample (see 3.2) shall be considered as compliance with this requirement.

3.10.2 Temperature (see 4.15). -

(a) Operating: Ambient temperature in the range of +150°F to -65°F, except that the Control Recorder C-577( )/GMD-1 is not required to operate below -40°F. (The 150°F temperature includes effect of sun-load.) Exposure at the high temperature extreme shall not exceed 4 hours, and at the low temperature extreme not to exceed 72 hours, at any one time.

(b) Nonoperating. - Exposure in the range of +160°F to -80°F; exposure at the high temperature extreme not to exceed 4 hours, and at the low temperature extreme not to exceed 24 hours, at any one time.

3.10.3 Relative humidity (see 4.16). - Up to 97 percent relative humidity for 20 hours; and exposure at 100 percent relative humidity, with condensation, for 4 hours.

3.10.4 Immersion (see 4.18). - Three feet of fresh water for 2 hours (cases only).

3.10.5 Rain (see 4.17). - Two inches of rainfall per hour, with wind at 20 miles per hour. Do not subject the Control Recorder C-577( )/GMD-1 to this test.

3.10.6 Bench-handling. - The equipment shall meet full specification performance with no physical damage except abrasion after being subjected to the tests of 4.19.

3.11 Preconditioning. - The equipment shall be capable of meeting the requirements herein, without subsequent processing, after subjection to the bounce preconditioning of 4.6. (Also see 4.5)

3.12 Workmanship. - The equipment shall be manufactured and assembled in accordance with requirement 9 of MIL-STD-454 and the applicable portions of the following paragraphs in MIL-P-11268:

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General requirements for plastic material and plastic parts.
Wiring and cabling, including:
Slack
Protection
Clearance
Splicing
Connections, general
Grounding, general
Shielding on wire and cable
Soldering and brazing
Cleaning of units
General requirements for securing of parts
Process for welding
Cleaning after welding
Controls
Self-tapping screws
Cleaning prior to welding

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. - Inspection shall be classified as follows:

(a) Preproduction inspection(does not include preparation for delivery). (See 4.3)

(b) Inspection covered by subsidiary documents. (See 4.4)

(c) Quality conformance inspection.

(1) Quality conformance inspection of equipment before preparation for delivery. (See 4.5)

(2) Quality conformance inspection of preparation for delivery. (See 4.21)

4.3 Preproduction inspection. - This inspection will be performed by the contractor unless otherwise specified in the contract. It shall consist of the preproduction inspection specified in the subsidiary documents covering the items listed in 4.4 and the inspection specified for group A (Table I), group B, group C (Table II) and performed in that order except the flight test 4.14 of Table II shall be last.

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4.4 Inspection covered by subsidiary documents. - The following shall be inspected under the applicable subsidiary documents as part of the inspection of equipment before preparation for delivery:

Item	Where required
Finish	3.4
Marking	3.5

4.5 Quality conformance inspection of equipment before preparation for delivery. - The contractor shall perform the inspection specified in 4.4 and 4.5.1 through 4.5.4. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. The government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition the government --at its discretion--may perform all or any part of the specified inspection, to verify the contractor's compliance with specified requirements. (See 6.7) Test equipment for government verification inspection shall be made available by the contractor. Each unit shall be preconditioned after final assembly. (See 3.11)

4.5.1 Group A inspection. - This inspection including sampling, shall conform to Table 1 and the inspection procedures of MIL-STD-105, using the general inspection levels. Group A inspection shall be performed in any order which is satisfactory to the government, except that the flight test 4.14 shall be last.

Table 1 - Group A Inspection

Inspection	Reqt Para.	Insp Para.	AQL Major	AQL Minor
Visual and Mechanical	3.12	4.7	4.0 dphu*	15.0 dphu*
Receiver R-301( )/GMD-1	3.12	4.7	2.5 dphu*	10.0 dphu*
Antenna Control C-578( )/GMD-1	3.12	4.7	2.5 dphu*	10.0 dphu*
Control Recorder C-577( )/GMD-1	3.12	4.7	1.5 dphu*	6.5 dphu*
Azimuth Unit	3.12	4.7	1.5 dphu*	6.5 dphu*
Elevation Unit	3.12	4.7	1.5 dphu*	6.5 dphu*
Receiver and Antenna Control Housing	3.12	4.7	1.0 dphu*	4.0 dphu*
Control Recorder Cover	3.12	4.7	1.0 dphu*	4.0 dphu*
Antenna reflector	3.12	4.7	1.0 dphu*	4.0 dphu*
Spinner Assembly	3.12	4.7		
Miscellaneous parts such as compression bars, outrigger, legs, transit cases, etc.	3.12	4.7	1%	4%
Cases	3.12	4.7	1%	4%

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Performance				
Mechanical	3.8.2	4.11	1%	4%
Tracking	3.8.1	4.10)	1%	**
Control Recorder C-577( )/GMD-1	3.8.3	4.14)	For The Group	
Antenna AS-462( )/GMD-1	3.8.4	4.12)		
Rawin Receiver R-301( )/GMD-1	3.8.5	4.13)		
Flight Test	3.9	4.14	1%	**

*dphu = defects per hundred units
** all electrical defects are considered major

4.5.2 Group B inspection. - This inspection, including sampling, shall conform to 3.6 and to the procedures for small-sample inspection of MIL-STD-105, using the special inspection levels. Group B inspection shall normally be performed on inspection lots that have passed Group A inspection and on samples selected from units that have been subjected to and met the Group A inspection. The AQL shall be 6.5 percent defective and the inspection level shall be S-4 for normal, tightened, and reduced inspection.

4.5.3 Group C inspection. - This inspection shall consist of the tests specified in Table II and shall be performed on sample units that have been subjected to and met Groups A and B inspection. Sample units shall be selected in accordance with paragraph 4.5.3.1

Table II - Group C Inspection

Inspection	Requirement Paragraph	Inspection Paragraph
Sub-Group I		
Temperature (Cases not included)	3.10.2	4.15
Humidity (Cases not included)	3.10.3	4.16
Immersion (Cases only)	3.10.4	4.18
Rain (Cases not included)	3.10.5	4.17
Sub-Group II		
Shock; bench-handling	3.10.6	4.19
Fungus	3.7.1	4.20.1
Salt Fog	3.7.2	4.20.2

Table IIA - Group C Inspection Test Sequence

Unit 1	Unit 2	Unit 3
Temperature	Humidity	Fungus
Shock; bench handling	Salt Fog	
Rain		
Immersion		

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4.5.3.1 Sampling for Inspection of Equipment. - Production sampling for Group C inspection shall be three (3) items selected from the first ten (10) items produced and subjected to Sub-Group 1 & 11 inspection. One (1) item for each twenty(20) produced thereafter shall be subjected to Sub-Group 1 inspection. The items shall be selected at random from each group of units to be subjected to Group C inspection. When preproduction samples are specified by contract or purchase order Group C inspection will not be required for three items of the first ten production units.

4.5.3.2 Noncompliance. - The contractor shall immediately report in writing each Group C failure occurrence, including details of the failure and characteristics affected. The contractor shall immediately investigate the cause of failure and further report the results of investigation and details of the proposed corrective action on (i) the process and materials, as applicable, and (ii) all units of product which were manufactured under the same conditions and which the government considers subject to the same failure. Reports shall be forwarded to the responsible technical activity designated in the contract through the Quality Assurance Representative. After corrective action has been taken, additional sample units shall be subjected to Group C inspection (all inspections, or the inspections which the sample failed, at the option of the government) and Groups A and B inspection may be reinstituted; however, final acceptance and shipment will be withhold until the Group C reinspection results have shown that the corrective action was effective.

4.5.4 Reinspection of Conforming Group C Sample Units. - Unless otherwise specified, sample units which have been subjected to and passed Group C inspection may be accepted on the contract or order provided all damage is repaired and the sample units are resubjected to and pass Group A inspection.

4.6 Bounce preconditioning. - The unit, with shock mounts (if any) removed or blocked, shall be placed in its normal operation position on the table of the Package Tester, Type 1000-SC, as made by L.A.B. Corporation, Skaneatales, New York, or equal. The package tester, shafts in phase, shall have a speed such that it is just possible to insert a 1/32-inch-thick strip of material under one corner or edge of the unit to a distance of 3 inches as the unit bounces. The unit shall be subjected to this preconditioning for 5 minutes. After bounce preconditioning, the unit shall not be repaired, alined, cleaned, or otherwise changed prior to subjection to acceptance inspection.

4.7 Visual and mechanical inspection. - The equipment shall be examined for the defects listed in MIL-STD-454.

4.7.1 Inspection for dimensional interchangeability. - The dimensions listed below shall be gaged or measured to determine conformance to the physical interchangeability requirement of 3.6. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

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(a) External and internal dimensions of cases, covers, and insertable assemblies, when such dimensions affect mating of parts.

(b) Dimensions of cavities, when such dimensions affect insertion of items.

(c) Location of hinges and fasteners on separable parts or assemblies which must mate, such as cases, covers, and mounting.

(d) Location of connectors, locking pins, fasteners, slides, and mountings which receive mating parts of plug-in assemblies and major units; and location of the mating parts on the plug-in assembly or major unit.

(e) Size and form of special threads.

4.8 Test equipment and power input. - Test equipment shall be calibrated in accordance with this specification. When the calibration is not specified MIL-C-45662 shall apply. The nominal input power of the AN/GMD-1( ) is 117 VAC at 60 Hz. (See 3.8)

4.9 Rawin Set AN/GMD-1( ), Tests and units of tests. - When either the Antenna AS-462( )/GMD-1( ), Antenna Control C-578( )/GMD-1, Control Recorder C-577( )/GMD-1, Pedestal AB-159( )/GMD-1 or the Rawin Receiver R-301( )/GMD-1 are procured as separate units, they may be tested as a part of the AN/GMD-1( ) or they may be tested with a test set-up that simulates the AN/GMD-1( ). All units shall also be subjected to all the additional applicable tests of this specification. The first test for a complete Rawin Set should be 4.11.

4.10 Tracking. - The following tests shall be performed at a suitable test field. A fixed target transmitter shall be positioned at a distance of 150 feet or more and at an elevation angle of at least 10° measured at the antenna. An adequate means of support for the transmitter shall be provided. In the tests shown in subprargarphs a, b, c, and d the indicated bearing shall conform to 3.8.1.1. During the tests compliance with 3.8.1.3 thru 3.8.1.6 shall be determined.

(a) Tuning of the Receiver. - With the receiver system locked in on the target and the target radiated power adjusted to produce a demodulated pulse that is two times the peak value of the peak noise as measured with a wide band oscilloscope or equal at the output of the I. F. strip, the automatic manual tuning of Rawin Receiver in the manual position, and the target transmitter adjusted to frequencies of 1660, and 1700 MHz in turn, the receiver frequency shall be adjusted plus and minus 2 MHz of the target frequency.

(b) Changes in field strength. - The output of the target transmitter at 1,680 MHz shall be varied from 400 milliwatts to the minimum level that permits tracking.

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(c) Changes in polarization. - With the transmitter adjusted as specified by (a), the antenna of the target transmitter shall be rotated 45° clockwise then 45° counterclock-wise from the vertical and checked at frequencies of 1,660, 1,680 and 1,700 MHz.

(d) Changes in carrier frequency. - With the transmitter adjusted as specified by (a), the carrier frequency shall be set in sequency at 1,660, 1,680 and 1,700 MHz and the change in bearing from 1,680 MHz point measured.

4.10.1 Azimuth - elevation automatic reset. - The test set up is the same as 4.10 (transmitter set at 1680 MHz and radiating 180 milliwatts) with each test performed five times and the target bearing shall be determined optically. Each of the following subparagraphs a, b, c and d shall conform to 3.8.1.2.

(a) The antenna shall then be taken at least 6° off bearing in a clock-wise direction by the action of the manual control. The antenna shall then reset itself automatically on the target.

(b) Repeat in a counterclock-wise direction.

(c) The antenna shall then be taken off target by at least 6° in the upward direction by the action of the manual control. The antenna shall then reset itself automatically on target.

(d) Repeat in the downward direction.

4.11 Mechanical. -

4.11.1 Azimuth gear test.- The telescope attached to the antenna shall be sighted on a target at a minimum distance of 100 yards. The antenna shall be rotated 3° in azimuth in a clockwise direction, as indicated on the remote azimuth indicator, and then returned by counterclock-wise rotation to the fixed point as sighted through the telescope, but not overrunning the fixed point. The reading of the indicator shall be recorded. The antenna shall then be rotated 3° in azimuth in a counterclock-wise direction and returned by clock-wise rotation until the fixed point is again sighted on the cross hairs of the telescope and the indicator reading recorded. This test shall be performed at least 5 times in each octant of azimuth and the average deviation shall conform to 3.8.2.

4.11.2 Elevation gear test. - The test specified in 4.11.1 shall be repeated for the elevation gear, with the antenna properly counter-balanced and the antenna rotated in elevation rather than azimuth at elevation angles of 10°, 30° and 60°. Each test shall be repeated 5 times and average of each test (10°, 30° and 60°) shall conform to 3.8.2.

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4.11.2.1 Alternate elevation gear test. - Elevation change (backlash) in the elevation unit gears, as measured as a displacement of a gear, shall be made at 10°, 30° and 60° of elevation and conform to 3.8.2. As it is impractical to have stationary objects close by and of sufficient height to sight on at these high angles of elevation, the following method of performing the elevation gear test is acceptable.

(a) With the loading springs disconnected, place the elevation unit face (surface on which the antenna is mounted) down on a bench. No power is to be used. All movement shall be done by hand.

(b) Refer to Drawing SC-D-86346.

(c) Locate on the drawing, and then in the elevation unit, the gear next to the motor gear. (454 rpm)

(d) Provide a stationary reference index to permit measurement of the displacement of this gear.

(e) Move the elevation yoke to a position equivalent to 10° (30°, 60°, 90°) in elevation.

(f) Move the 454 rpm (143 teeth) gear counter-clockwise 1/4 turn, and with a pencil or crayon, make reference mark on the tooth of this gear that is adjacent to the stationary reference index ((d) above).

(g) Slowly move the 454 rpm (143 teeth) gear clockwise until the elevation yoke just starts to move.

(h) Count the teeth on the 454-rpm (143 teeth) gear that were moved past the referenced marked index. The number of teeth passed shall not exceed five.

(i) Erase the reference mark on the 454-rpm (143 teeth) gear ((f) above) and repeat the test by moving this gear 1/4 turn clockwise from the equivalent 10° (30°, 60°) elevation position, and then slowly move the 454 rpm (143 teeth) gear counter-clockwise until the elevation yoke just starts to move.

(j) Likewise check the backlash, both counter-clockwise and clockwise for 30° and 60° respectively.

4.12 Antenna AS-462( )/GRD-1. - By the use of a crystal detector or a bolometer with a suitably calibrated output, the following measurements shall be taken:

(a) The width of the antenna directivity pattern shall conform to 3.8.4(a).

(b) Maximum amplitude of secondary lobes shall conform to 3.8.4(b).

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(c) Intersection of the two azimuth directivity patterns shall conform to 3.8.4(c).

(d) Intersection of the two elevation directivity patterns shall conform to 3.8.4(c).

4.13. Receiver R-301( )/GRD-1 tests. -

4.13.1 Frequency. - The frequency range shall be checked for conformance to 3.8.5.1.

4.13.2 IF amplifier. - The overall sensitivity of the receiver and the shape of the IF band-width curve shall conform to 3.8.5.2 by the use of an accurate signal generator and oscilloscope, or equal.

4.13.3 Automatic-frequency correction (AFC). - Performance of the automatic frequency correction (AFC) shall be tested for conformance to 3.8.5.3.

4.13.4, Automatic gain control (AGC). - Test the AGC for conformance to 3.8.5.4.

4.13.5 Audio channel phase shift . - Vary the frequency of the power supply from its normal frequency to a low limit of 50 Hz then to the high limit of 65 Hz and vary the input signal level from 5 microvolts to 50,000 microvolts separately and in combination. The maximum change in phase shift shall conform to 3.8.5.5.

4.13.6 Signal-to-noise ratio. - The signal-to-noise ratio shall be measured as follows and conform to 3.8.5.6:

(a) Connect a noise diode (such as a Mega-Node) through the mixer to the receiver.

(b) Connect a microammeter in series with a 68,000-ohms resistor to unloaded short AF output of the receiver.

(c) Record the meter reading.

(d) Turn on the noise diode and increase its output so as to increase the meter reading to 1.41 times the reading in (c).

(e) Record the noise diode reading in db.

(f) Apply any correction factor which is characteristic of the noise diode.

(g) Add 3 db to the result of (f).

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4.13.7 Radiosonde pulse channel. - The performance of the radiosonde pulse channel shall be tested at 1,660, 1,680, and 1,700 Mc as described in the following subparagraphs and conform to 3.8.5.7.

4.13.7.1 Low level performance. - Connect Test Set TS-538( )/U, whose pulse width has been adjusted to 4.40 microsecond, to the input of the receiver. Connect an electronic frequency counter, or equal, to the output of the receiver. Operate the attenuator of the test set so that the signal pulse peaks are twice the noise peaks at the output of the IF amplifier as measured on a broad-band oscilloscope or equal. The electronic frequency counter shall steadily and accurately indicate the frequency of the input signal over the range of 10 to 225 Kc.

4.13.7.2 Positive pulse rejection. - With the receiver and test equipment connected and adjusted as in 4.13.7.1, connect a pulse generator, through a suitable compensated decoupling network, in parallel with the IF output of the receiver. Raise the output of Test Set TS-538( )/U to 6 dbm above the setting mentioned in 4.13.7.1. Adjust the level of the pulse generator so that it generates interfering positive pulses of one volt peak amplitude and 50 per second pulse rate. Adjust the repetition rate of the test set to 200 Hz. Increase the width of the interfering pulses from one microsecond to the width which first causes instability of the frequency counter. This width shall not be less than 20 micro-seconds. The preceding test shall be repeated with the test set adjusted to repetition rates of 100, 20 and 10 Hz and the pulse generator to repetition rates of 500 and 2,000 Hz.

4.13.7.3 Negative pulse rejection. - The tests specified in paragraph 4.13.7.2 shall be repeated except that the pulse generator shall be adjusted to generate negative pulses 10 times to peak amplitude of the pulses from the TS-533( )/U.

4.13.7.4 Output pulse. - The width and amplitude of the output pulses of the receiver as applied to an oscilloscope or equal be measured simulating the connecting cable specified in 3.8.5.7 by a 0.01 mfd capacitor across the output.

4.13.8 Audio monitor. - The audio monitor shall conform to 3.8.5.8.

4.13.9 Microphonics. - The receiver shall be operated with a 1,680 Mc signal at 50 microvolts input modulated by pulses of 200 Hz as derived from Test Set TS-538( )/U. When the tubes are tapped, the signal shall conform to 3.8.5.9.

4.13.10 Mixer, impedance. - The mixer shall be tested for conformance with 3.8.5.10.

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4.14 Flight test. - The preproduction samples or the first production units specified by 3.2 and 4.5.3.1 shall be subjected to at least 10 flight tests and each production unit to at least one flight test of the tests specified below:

NOTE: Radiosondes, Balloons and String will be government furnished.

The antenna shall be aimed at the flight transmitter as it is about to be released. Upon release and for as long a time thereafter as the input level to the mixer exceeds 5 microvolts or until the balloon bursts. The accuracy shall be judged by noting the position of the balloon in the field of the telescope (M-17) mounted to the antenna structure. Readings shall be taken at regular intervals of not more than 15 seconds to determine compliance with 3.9.

4.15 Temperature test. - The equipment shall be subjected to the temperature cycle shown on MIL-STD-169 and the measurements taken at steps 1,3,7 and 10. The performance shall conform to 4.16.3. The Control Recorder C-577( )/GMD-1( ) operation is not required below -40°F.

4.16 Moisture-resistance test. - The equipment shall be tested as follows:

4.16.1 Test conditions. -

(a) Do not move equipment from the humidity chamber for measurements.

(b) Start measurements not more than 5 minutes after power is applied to the equipment. Complete measurements as rapidly as possible. Do not leave power on after measurements have been completed.

(c) Test sealed units with the seal intact during last two cycles. Bring out leads for electrical measurements, through a hole in the case and seal against entrance of moisture. Perform last two cycles with sealed case removed.

4.16.2 Test method. -

(a) Dry at 130° ±5°F for 24 hours.

(b) Condition at 77° ±5°F and 40 to 50 percent relative humidity for 24 hours.

(c) Take measurements as specified in 4.16.3 and readjust or realine as necessary to meet full specification requirements.

(d) Subject to continuous cycling for five 48-hour cycles. Temperature, relative humidity, and period of time for each portion of the cycle shall conform to MIL-STD-170. The measurements specified in 4.16.3 shall be made during the periods shown on the standard and shall comply with 4.16.3.

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(e) After cycling has been completed, condition the equipment for 24 hours at 77° ±5°F and 40 to 60 percent relative humidity. Then adjust for optimum performance, using only those means provided by the equipment. No repair or replacement of parts shall be made. After adjustment, the equipment shall meet full specified performance for those measurements specified in 4.16.3.

4.16.3 Performance. - Measurements made during test, and specified performance during cycling shall be as follows:

(a) Complete set. - The complete set shall demonstrate capability of automatic tracking and rotation of at least 1 rpm in azimuth and elevation when manual tracking controls are operated.

(b) Receiver. - Receiver shall demonstrate ability to reproduce modulation of a 5 microvolt signal from TS-538( )/U in its loud-speaker. Both meters shall show correct indications. The service meter shall show correct indications in all switch positions. AFC performance shall be in accordance with 3.8.5.3.

(c) Control Recorder C-577( )/GMD-1. - The control recorder shall conform to 3.8.3.

4.16.4 Failure. - If any unit fails to meet full specified performance during cycling, it shall be realined or readjusted once. If the unit then fails to meet 4.16.3 or fails subsequently during cycling, it does not pass the test. In addition, if the unit fails to meet full specification requirements, as specified in 4.16.2(e), it does not pass the test.

4.17 Rain test. - The Rewin Set AN/GMD-1( ) assembled for operation except for the Control Recorder C-577( )/GMD-1 shall be tested as follows:

4.17.1 Dry at 150° ±5°F for 48 hours.

4.17.2 Condition for 4 hours at 77° ±5°F and 40 to 60 percent relative humidity.

4.17.3 Take the measurements in 4.16.3 to insure that the equipment meets specified performance.

4.17.4 Place the equipment, in its normal operating position, in a test chamber maintained at 77° ±5°F. Allow simulated rainfall to fall on exposed surfaces of the equipment. The water shall be at a temperature of 65° ±5°F and shall fall freely at a rate of 2 inches per hour. Maintain this condition for 1 hour.

4.17.5 With the rainfall continuing at the 2-inch rate, introduce a simulated wind blowing horizontally at 20 ±2 miles per hour against one side of the equipment. The water shall fall on exposed surfaces and be blown by the wind against one side of the equipment. Maintain this condition for 1 hour.

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4.17.6 Rotate the equipment 90 degrees horizontally, to expose a second side of the component to the wind and rain. Maintain this condition for 1 hour.

4.17.7 Continue as in 4.17.6, until all four sides of the equipment have been exposed to the wind and rain, each side for 1 hour.

4.17.8 Immediately after completion of the foregoing, repeat the measurements specified in 4.17.3 without removal from the chamber but with the rain and wind discontinued. Complete the measurements within 1 hour. The equipment shall meet specified performance.

4.17.9 Open the equipment. There shall be no visible evidence of leakage.

4.18 Immersion test (cases only). - The cases, as prepared for field transportation, shall be immersed to a minimum depth of 3 feet of fresh water for 2 hours. Immediately prior to immersion, the temperature of the equipment shall be 40°F, or more, above the temperature of the water. The tank in which the equipment is immersed shall be of sufficient capacity to maintain the water within +20°F of its initial temperature, or the temperature of the water shall be maintained within those limits by other means. After completion of the 2-hour period of immersion, the equipment shall be removed from the water and wiped dry on exterior surfaces. When the cases are opened, there shall be no evidence of leakage.

4.19. Shock test: bench-handling (Antenna Control C-578( )/GMD-1, Control Recorder C-577( )/GMD-1 and Rawin Receiver R-301( )/GMD-1 only). - The chasis-and-front-panel assembly shall be removed from its enclosure, as for servicing, and placed in a suitable position for servicing on a solid, 2-inch fir bench top. The test shall be performed as follows, in a manner simulating shocks liable to occur during servicing:

(a) Tilt up the assembly through an angle of 30 degrees, using one edge of the assembly as an axis, and permit the assembly to drop back freely to the horizontal. Repeat, using other practicable edges of the same horizontaly face as axis, for a total of four drops.

(b) Repeat "(a)", with the assembly resting on other faces, until it has been dropped for a total of four times on each face on which the assembly might practically be placed during servicing.

4.20 Tropicalization. -

4.20.1 Fungus test. - The equipment (including accessories) shall be tested in accordance with Method 508, Procedure 1, of MIL-STD-810. Spraying shall include parts and materials on and under the chassis, and the interior of the equipment case. The equipment shall remain in an open condition during the test. Operation is not required.

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4.20.2 Salt Fog. - The equipment and assemblies shall be exposed in their operating configuration to salt fog in accordance with Method 509 of MIL-STD-810 for 48 hours. Equipment and assemblies shall not be opened and shall not be required to be operated.

4.21 Quality conformance inspection of preparation for delivery. - Preparation for delivery shall be inspected in accordance with MIL-P-116 to determine conformance to the requirement of Section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. - Preservation and packaging shall be level A or C, as specified (see 6.2).

5.1.1 Leval A

5.1.1.1 Cleaning. - Rawin Set AN/GMD-1 shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. - Rawin Set AN/GMD-1 shall be dried in accordance with the applicable procedure of MIL-P-116.

5.1.1.3 Preservation application. None required

5.1.1.4 Unit packaging. - Unit packaging shall be in accordance with the methods prescribed in MIL-P-116, as specified in figures 1 thru 6 and tables 1 thru XI.

5.1.2 Level C. Each Rawin Set AN/GMD-1 shall be preserved and packaged in a manner that will afford adequate protection against corrosion, deterioration and damage during shipment from the supply source to the first receiving activity.

5.2 Packing. - Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Level A. - Each Rawin Set AN/GMD-1( ), packaged as specified in 5.1, shall be packed as specified in figures 1 thru 6 and tables 1 thru XI.

5.2.2 Level B. - Each Rawin Set AN/GMD-1( ), packaged as specified in 5.1, shall be packed as specified in 5.2.1, except that the wood boxes shall be class 1.

5.2.3 Level C. - Each Rawin Set AN/GMD-1( ), packaged as specified in 5-1, shall be packed in shipping containers in a manner that will afford adequate protection against damage to the package and its contents during shipment from the supply source to the first receiving activity. Shipping containers shall comply with the rules and regulations of the common carrier as applicable to the mode of transportation.

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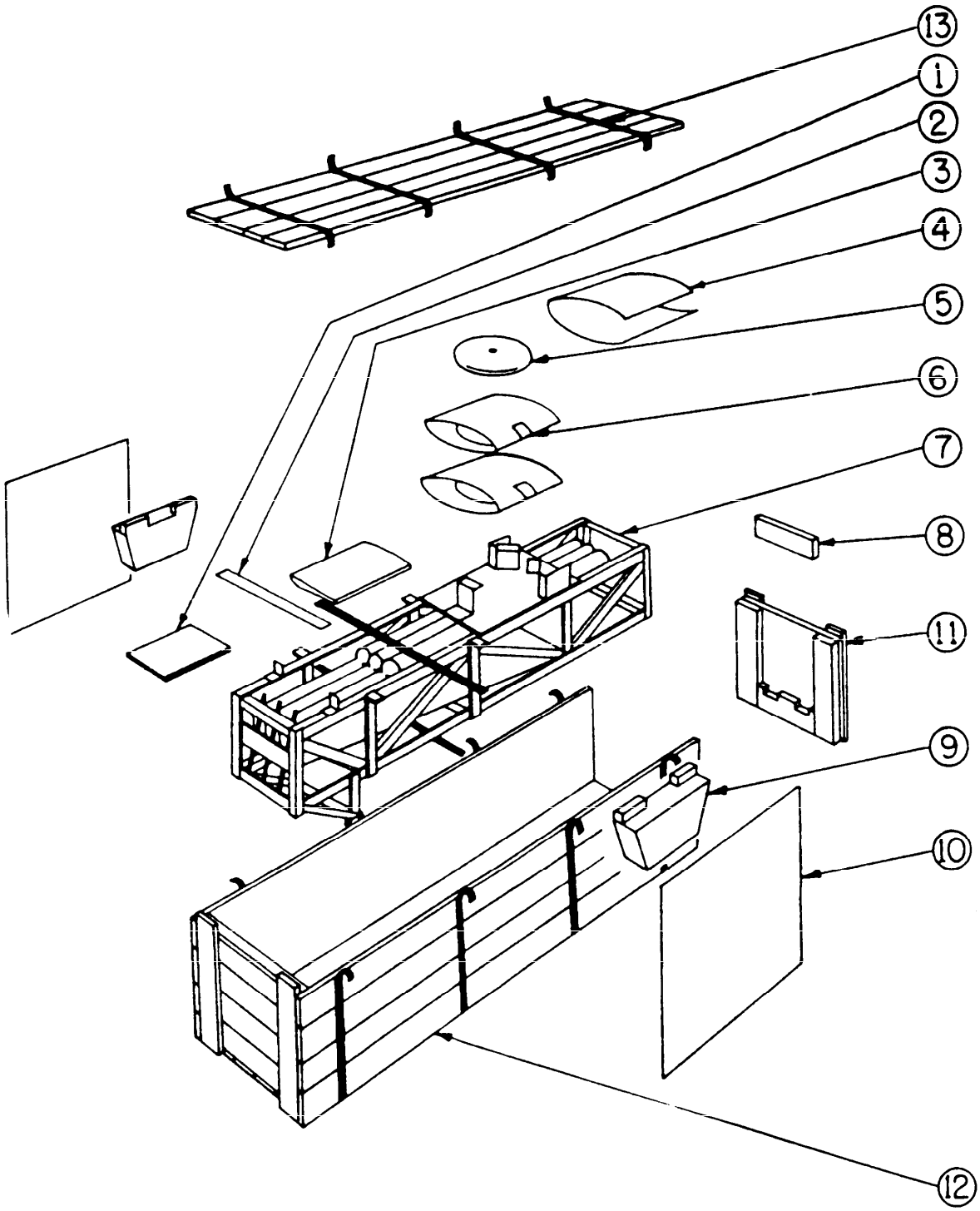


FIGURE 1

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BILL OF MATERIAL
TABLE I
Box 1 of 9

REF No.	ITEM Nomenclature	No. Reqd	Method MIL-P-116	SIZE - INCHES L	W	D	REQUIREMENT Spec No.	T	C	V	G	F	S
1	Tech Lit	2	IC-1	11-1/2	9		MIL-B-117	II	C				
2	Cable, Guy Ay	4	III										
	Guy	3	III										
	Leg, Tripod Ass	3	III										
	Support, Leg	3	III										
3	Wrap	13	as required				PPP-P-291	III					
	Tape	13	as required				PPP-T-60	IV					
4	Wrap	3		12	16		PPP-P-291	III					
5	Plate, Tripod	3	III										
6	Tape	6		12	2		PPP-T-60	IV					
7	Leg, Pack Frm	1	III										
8	Block, Wood	2					PPP-B-621						
9	Support, Wood	2	III										
10	Wrap	2					UU-P-268				B	30 wt	
11	Box, Ends	2					PPP-B-621						
12	Box, Wood	1		71-3/4	24-3/4	22-3/4	PPP-B-621		2				4
	Caseliner	1					MIL-L-10547				B		
				All items will be immobilized within the box by the use of wood blocks, braces or cleats as required.									
13	Strapping	4		110	3/4	.023	QQ-S-781	1	B				

Lev. A - PACKED WEIGHT 250 lbs
Lev. A - PACKED VOLUME 25.8 cu ft

T - Type V - Variety F - Flute
C - Class G - Grade S - Style

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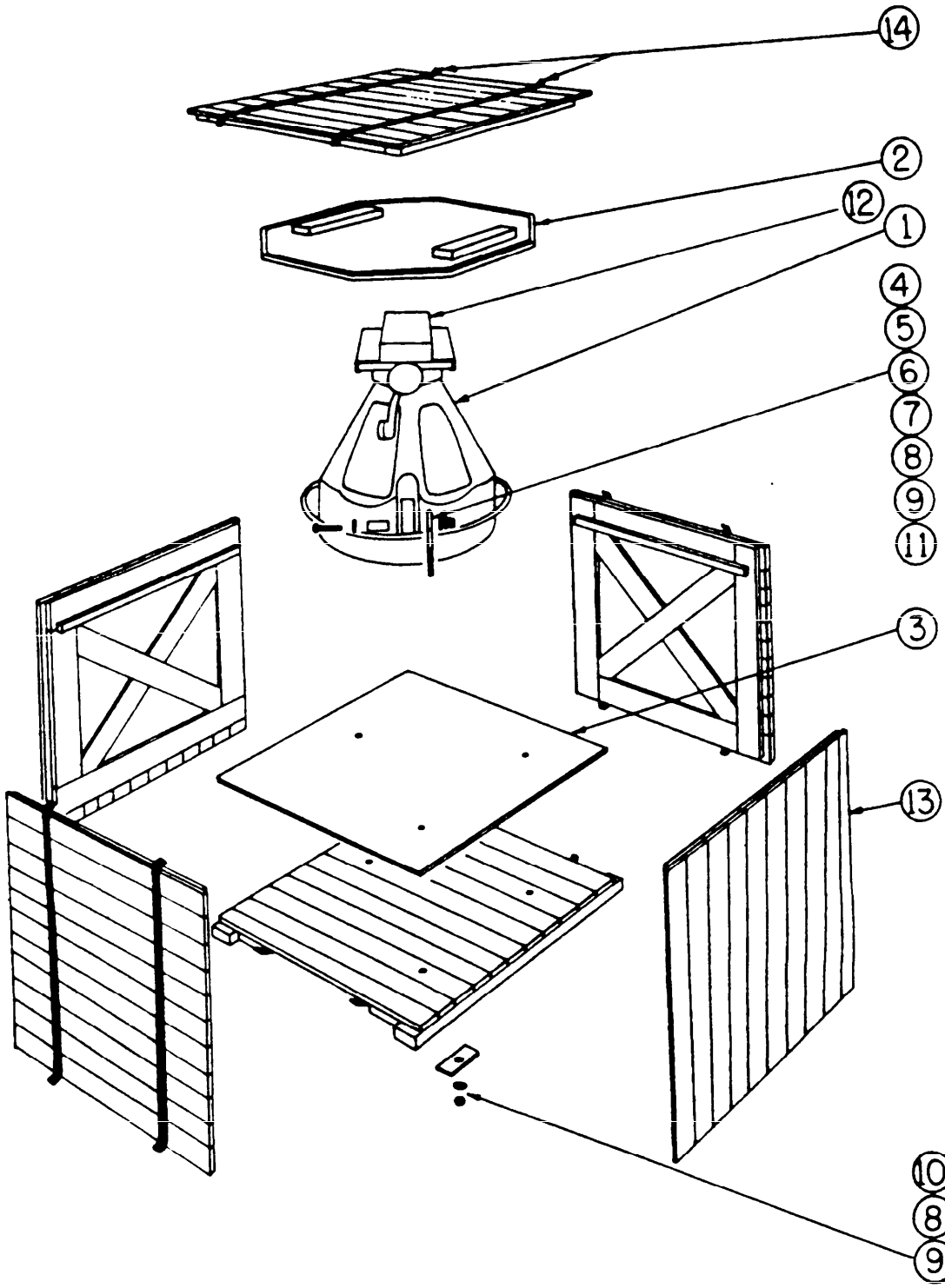


FIGURE 2

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BILL OF MATERIAL
TABLE II
Box 2 of 9

REF No.	ITEM Nomenclature	No. Reqd	Method MIL-P-116	SIZE - INCHES L	W	D	REQUIREMENT Spec No.	T	C	V	G	F	S
1	Azimuth Unit	1	III										
2	Azimuth skid	1	III										
3	Cushioning	1		34-1/4	32-3/4		PPP-F-320	SF	WR		W5S		
4	Bolt Eye	3		9	3/8								
5	Insert Alm	3		1-3/8	3/4								
6	Nut, Steel	3											
7	Bolt, Steel	3		2	3/8								
8	Washer Flat	9											
9	Nut, Steel	3											
10	Plate, Steel	3		2	2	1/8							
11	Washer, Lock	6											
12	Azimuth Cover	1	III										
13	Box Wood	1		34-1/2	34-1/2	35-1/8	PPP-B-621		2				7
14	Strapping	3		150	3/4	.023	QQ-S-781	1	B				

Lev. A - PACKED WEIGHT 490 lbs
Lev. A - PACKED VOLUME 26 cu ft

T - Type V - Variety F - Flute
C - Class G - Grade S - Style

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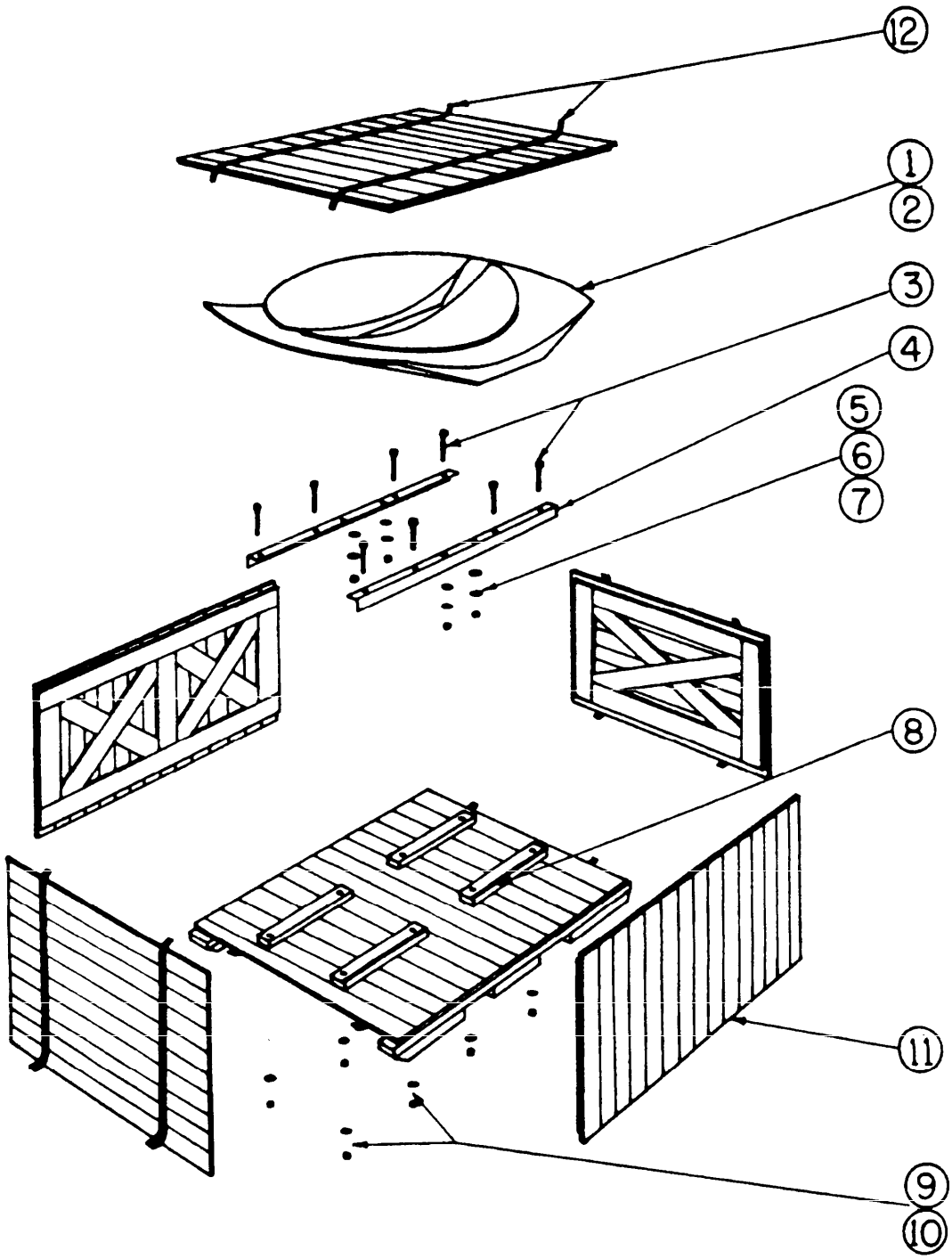


FIGURE 3

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BILL OF MATERIAL
TABLE III
Box 3 of 9

REF No.	ITEM Nomenclature	No. Reqd	Method MIL-P-116	SIZE - INCHES L	W	D	REQUIREMENT Spec No.	T	C	V	G	F	S
1	Reflector	1	III										
2	Wing Frame	1	III										
3	Bolts	8		3-1/4	3/8								
4	Shipping Mount	2		76-1/2	2	2							
5	Washer	4											
6	Lock Washer	4											
7	Nut, Hex	4											
8	Blocks, Wood	4		30	4	2	PPP-B-621						
9	Washer, 3/8	8											
10	Nut Hex, 3/8	8											
11	Box, Wood	1		91-3/4	53-1/4	32-1/8	PPP-B-621		2				7
12	Strapping	3		240	3/4	.023	QQ-S-781	1	B				

Lev. A - PACKED WEIGHT 440 lbs.
Lev. A - PACKED VOLUME 96 cu ft

T - Type
C - Class
V - Variety
G - Grade
F - Flute
S - Style

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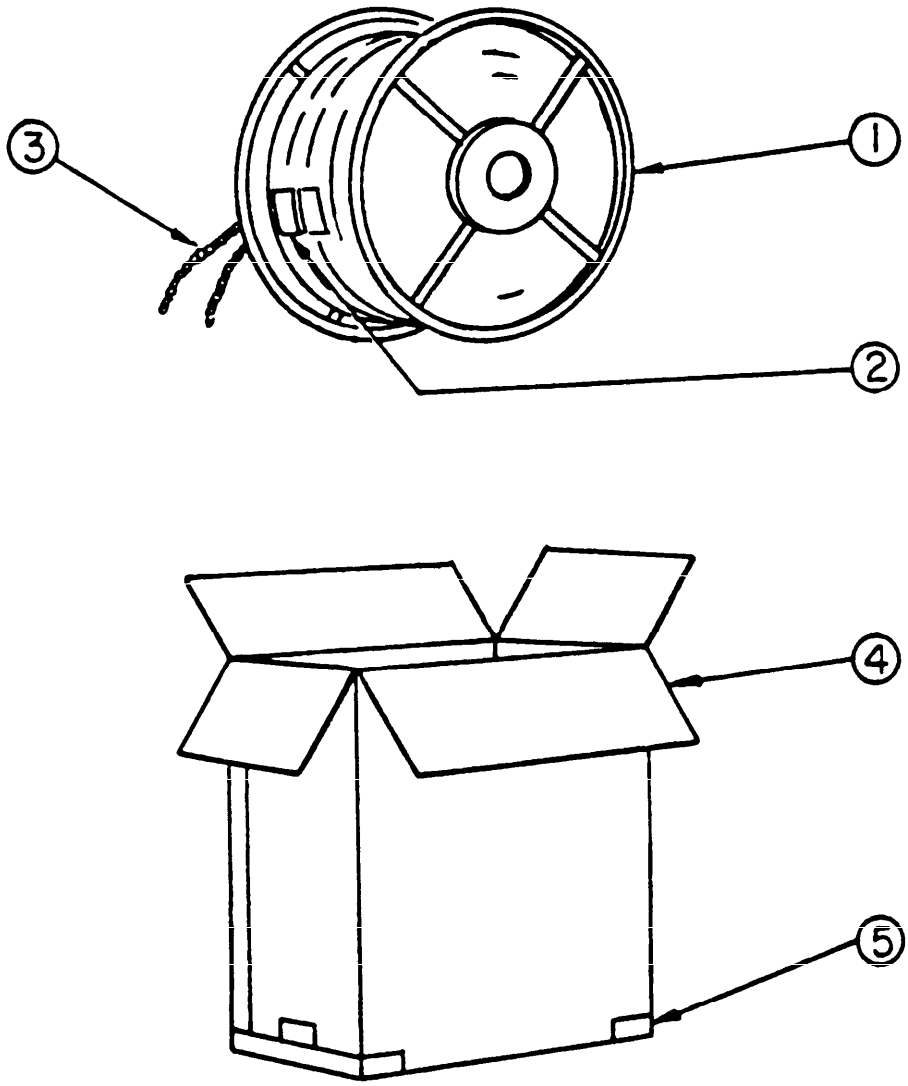


FIGURE 4

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BILL OF MATERIAL
TABLE IV
Box 4 of 9 (Partial)

REF No.	ITEM Nomenclature	No. Read	Method MIL-P-116	SIZE - INCHES L	W	D	REQUIREMENT Spec No.	T	C	V	G	F	S
1	Cable Reel RL-137( )/ GMD-1	1	III										
2	Cable Assy CX-1216( )/U	1	III										
3	Twine	1		18			T-T-871	II	1				
4	Box, Fiberboard	1		28	13-3/4	28	PPP-B-636	CF	WR	SW	W5c		RSC
5	Tape	4		18	2		PPP-T-76						
		2		34	2		PPP-T-76						
	FOR PACKING: SEE FIGURE 5 AND TABLE V. THIS ITEM PACKED TOGETHER WITH RL-138( )/GMD-1												

Lev. A - PACKED WEIGHT
Lev. A - PACKED VOLUME

T - Type
C - Class
V - Variety
G - Grade
F - Flute
S - Style

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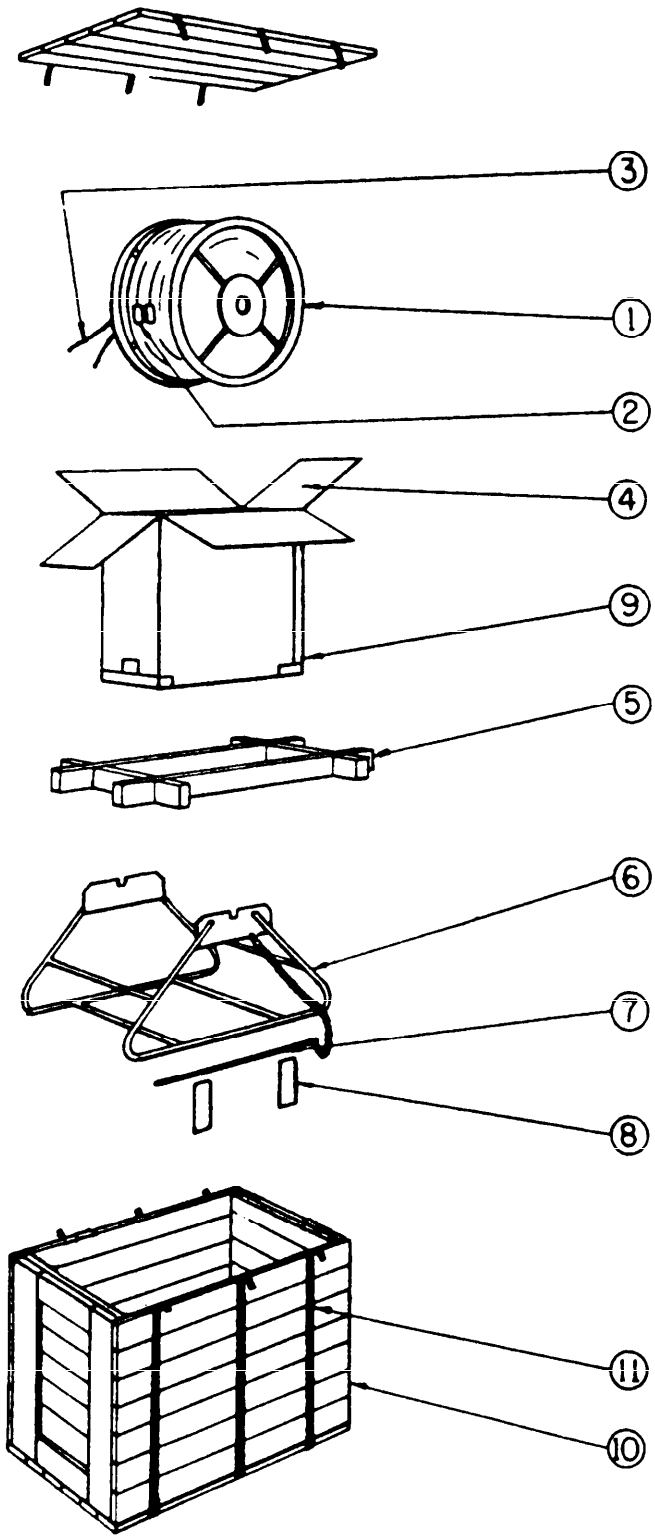


FIGURE 5

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BILL OF MATERIAL
TABLE V
Box 4 of 9 (Complete)

REF No.	ITEM Nomenclature	No. Req'd	Method MIL-P-116	SIZE - INCHES L	W	D	REQUIREMENT Spec No.	T	C	V	G	F	S
1	Cable Reel RL-138( )/ GMD-1	1	III										
2	Cable Assy CX-2043( )/U	1	III										
3	Twine	1		18			T-T-871	II	1				
4	Box, Fiberboard	1		22	7	22	PPP-B-636	CF	WR	SW	W5c		RSC
5	Brace, Wood 2 x 4	1					PPP-B-621						
6	Holder, Cable Reel MT-1421 ( )/U	1	III										
7	Rod Assembly	1											
8	Tape	2		4	2		PPP-T-60	IV					
9	Tape	4		13	2		PPP-T-76						
		2		28	2		PPP-T-76						
10	Box Wood	1		51	18	30	PPP-B-621		2				4
11	Strapping	3		100	3/4	.023	QQ-S-781	1	B				
	THIS ITEM PACKED TOGETHER WITH RL-137( )/GMD-1. SEE TABLE IV.												

Lev. A - PACKED WEIGHT 200 lbs
Lev. A - PACKED VOLUME 15.9 cu ft

T - Type
C - Class
V - Variety
G - Grade
F - Flute
S - Style

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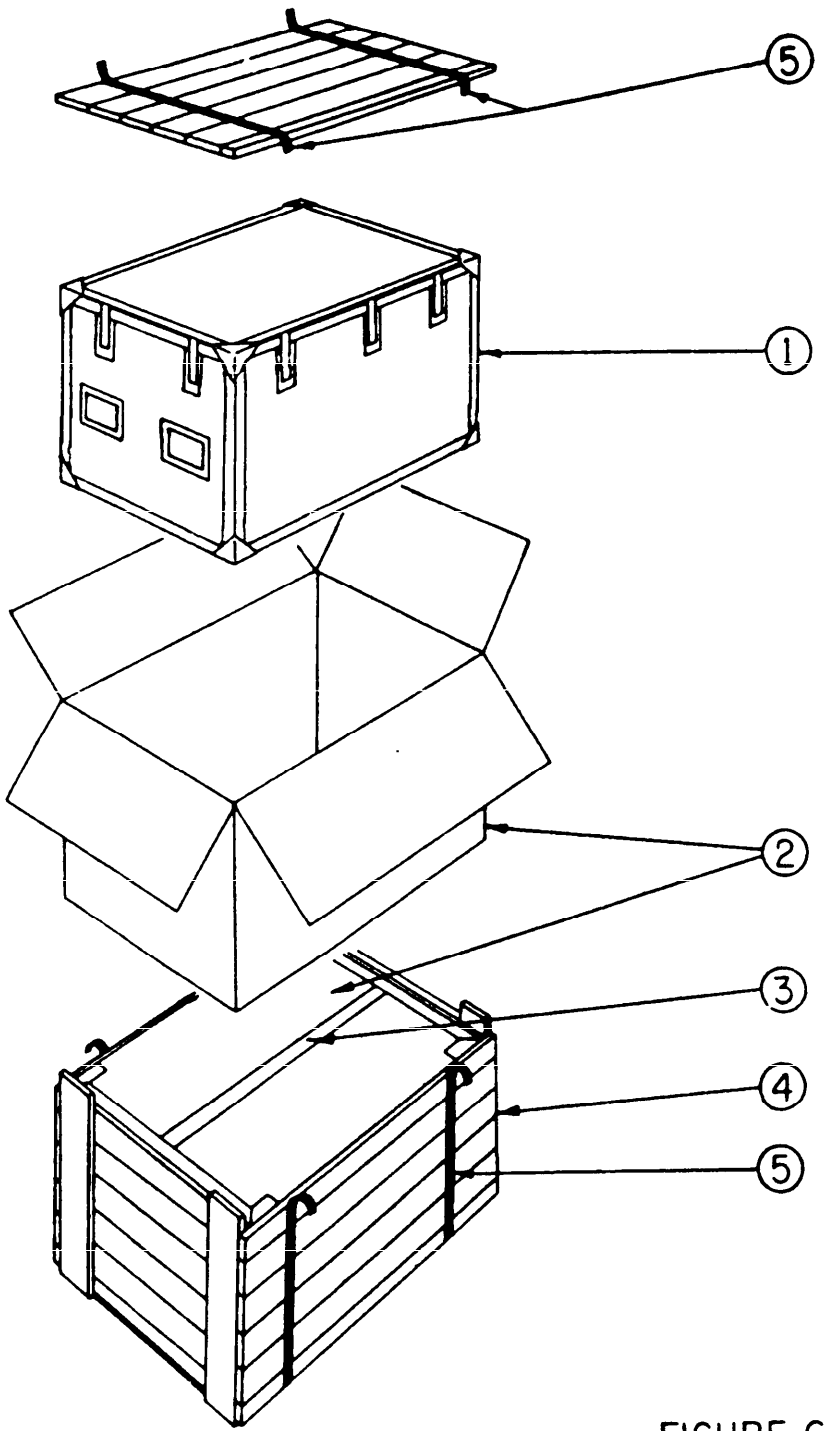


FIGURE 6

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BILL OF MATERIAL
TABLE VII
Box 6 of 9 (Partial)

REF No.	ITEM Nomenclature	No. Req'd	Method MIL-P-116	SIZE - INCHES L	W	D	REQUIREMENT Spec No.	T	C	V	G	F	S
1	Case CY-737( )/ GMD-1 includes: C-577( )/ GMD-1	1	III										
2	Box, Fiberboard	1		25-3/4	19	16-3/8	PPP-B-636	CF	WR	SW	W5c		RSC
3	Tape	4		25	2		PPP-T-76						
		2		32	2		PPP-T-76						
	FOR PACKING: SEE FIGURE 6 AND TABLE VIII. THIS ITEM PACKED TOGETHER WITH CASE CY-1895( )/GMD-1												

Lev. A - PACKED WEIGHT
Lev. A - PACKED VOLUME

T - Type
C - Class
V - Variety
G - Grade
F - Flute
S - Style

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BILL OF MATERIAL
TABLE VIII
Box 6 of 9 (Complete)

REF No.	ITEM Nomenclature	No. Reqd	Method MIL-P-116	SIZE - INCHES L	W	D	REQUIREMENT Spec No.	T	C	V	G	F	S
1	Case CY-1895( )/GMD-1 includes: Spare Parts (and Accessories) shall be packaged Method III in accordance with MIL-P-116. Electron Tubes shall be packaged in accordance with MIL-E-75.	1	III										
2	Box, Fiberboard	1		25-1/2	20-7/8	20-5/8	PPP-B-636	CF	WR	SW	W5c		RSC
3	Tape	4		27	2		PPP-T-76						
		2		32	2		PPP-T-76						
4	Box Wood	1		52	21-3/8	21-3/8	PPP-B-621		2				4
5	Strapping	2		96	3/4	.023	QQ-S-781	1	B				
	THIS ITEM PACKED TOGETHER WITH CASE CY-737( )/GMD-1. SEE TABLE VII.												

Lev. A - PACKED WEIGHT 200 lbs
Lev. A - PACKED VOLUME 13.3 cu ft

T - Type
C - Class
V - Variety
G - Grade
F - Flute
S - Style

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BILL OF MATERIAL
TABLE IX
Box 7 of 9

REF No.	ITEM Nomenclature	No. Req'd	Method MIL-P-116	SIZE - INCHES L	W	D	REQUIREMENT Spec No.	T	C	V	G	F	S
1	Case CY-735( )/ GMD-1 includes: Spare C-578( )/ GMD-1 R-301( )/ GMD-1	1	III										
2	Box, Fiberboard	1		25-3/8	24-5/8	20-3/8	PPP-B-636	CF	WR	SW	W5c		RSC
3	Tape	4		31	2		PPP-T-76						
		2		32	2		PPP-T-76						
4	Box, Wood	1		25-7/8	25-1/8	21-1/8	PPP-B-621		2				4
5	Strapping	2		96	3/4	.023	QQ-S-781	1	B				

Lev. A - PACKED WEIGHT 200 lbs
Lev. A - PACKED VOLUME 9 cu ft

T - Type
C - Class
V - Variety
G - Grade
F - Flute
S - Style

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BILL OF MATERIAL
TABLE X
Box 8 of 9

REF No.	ITEM Nomenclature	No. Read	Method MIL-P-116	SIZE - INCHES L	W	D	REQUIREMENT Spec No.	T	C	V	G	F	S
1	Case CY-734( )/ GMD-1 includes: Elevation Unit Yoke	1	III										
2	Box, Fiberboard	1		37-5/8	22-7/8	21-1/8	PPP-B-636	CF	WR	SW	W5c		RSC
3	Tape	4		29	2		PPP-T-76						
		2		44	2		PPP-T-76						
4	Box Wood	1		38-1/8	23-3/8	21-7/8	PPP-B-621		2				4
5	Strapping	2		108	3/4	.023	QQ-S-781	1	B				

Lev. A - PACKED WEIGHT 250 lbs
Lev. A - PACKED VOLUME 12.6 cu ft

T - Type
C - Class
V - Variety
G - Grade
F - Flute
S - Style

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BILL OF MATERIAL
TABLE VI
Box 5 of 9

REF No.	ITEM Nomenclature	No. Req'd	Method MIL-P-116	SIZE - INCHES L	W	D	REQUIREMENT Spec No.	T	C	V	G	F	S
1	Case CY-736( )/ GMD-1 includes: C-578( )/ GMD-1 Housing R-301( )/ GMD-1	1	III										
2	Box, Fiberboard	1		32-1/2	26-3/8	28-1/4	PPP-B-636	CF	WR	SW	W5c		RSC
3	Tape	4		33	2		PPP-T-76						
		2		39	2		PPP-T-76						
4	Box, Wood	1		33	26-7/8	29	PPP-B-621		2				4
5	Strapping	2		120	3/4	.023	QQ-S-781	1	B				

Lev. A - PACKED WEIGHT 350 lbs
Lev. A - PACKED VOLUME 16.4 cu ft

T - Type V - Variety F - Flute
C - Class G - Grade S - Style

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BILL OF MATERIAL
TABLE XI
Box 9 of 9

REF No.	ITEM Nomenclature	No. Req'd	Method MIL-P-116	SIZE - INCHES L	W	D	REQUIREMENT Spec No.	T	C	V	G	F	S
1	Case CY-1157( )/GMD-1 includes:	1	III										
	Minor components:												
	Cable Assys												
	Connector												
	Grd rod												
	M-17 (GFE)												
	TS-538( )/U												
	Tool												
	Wrenches												
	etc.												
2	Box, Fiberboard	1		45	26-1/4	15-5/8	PPP-B-636	CF	WR	SW	W5c		RSC
3	Tape	4		33	2		PPP-T-76						
		2		51	2		PPP-T-76						
4	Box Wood	1											
5	Strapping	2		90	3/4	.023	QQ-S-781	1	B				

Lev. A - PACKED WEIGHT 250 lbs
Lev. A - PACKED VOLUME 13 cu ft

T - Type
C - Class
V - Variety
G - Grade
F - Flute
S - Style

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5.3 Marking. - In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6 NOTES

6.1 Intended use. - Rawin Set AN/GMD-1( ) is a transportable ground-installed radio direction finder used for automatically tracing a balloon-borne radiosonde transmitter. From azimuth and elevation angles thus determine the speed and direction of the wind at various altitudes can be calculated. This equipment also receives and delivers to a supplementary recorder intelligence of upper-strata meteoroloical conditions as transmitted by the radiosonde.

6.2 Ordering data. - Procurement documents should specify the following:

(a) Title, number, and date of this specification and any amendment thereto.

(b) Type required.

(c) Level of preservation, packaging, packing and marking required. (See Section 5)

(d) The specific paragraphs of section 5 which are applicable to the particular procurement.

(e) Marking and shipping of samples.

(f) Place of final inspection

(g) Submission of the statement of treatment should be submitted to the contracting officer as soon as possible after award of contract.

(h) Balloons, Radiosondes and String to be supplied. (See 4.14)

6.3 Dimensional data. - Dimensions of packaging materials in the Bill of Materials Table are based on the dimensions of the items cited in the Table. When the dimensions of the items vary from those cited, the dimensions of the packaging materials shall be adjusted accordingly.

6.4 Production Inspection: - Three sample units of each item cited in section 1 are generally required so that lengthy environmental tests can be completed on two sample units while complete performance measurements can be made on the third sample unit. (See 3.?)

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6.5 Group C inspection. - Approval to ship may be withheld, at the discretion of the government, pending the decision from the contracting officer on the adequacy of corrective action. (See 4.5.3.2)

6.6 Nomenclature. - The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example: AN/GMD-1[ ]. The contractor should apply for nomenclature in accordance with the applicable clause in the contract. (See 1.1)

6.7 Verification inspection. - Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

Custodians:

Army - EL
Navy - AS
Air Force - 01

Preparing Activity:

Army - EL

Project Number 6660-0001

Reviewer Activities:

Army - ME
Air Force - 71
Navy - AS

User Activities:

Army - MI
Navy - AS
Navy - MC

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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